CIRC HQ STUDENT HANDOUT | GRADES 5 AND 6

Hull Design

Goal: Shape a foil hull that stays stable while carrying the greatest tested payload.

35 minutes | 8 steps | Engineering and Design | Playbook B meeting 7

Ask | Plan | Build | Test | Improve

You are done when

Record the best cargo count and explain how hull shape affected stability.

You will use

- Aluminum foil sheets
- Water tubs
- Uniform cargo pieces
- Towels
- Recording sheet

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 3 min

- Place one foil sheet beside the tub.
- Count the cargo pieces.

2 Step 2: Water safety — 3 min

- Keep water inside the tub.
- Wipe each spill with a towel.
- Keep electronics outside the wet zone.

3 Step 3: Plan the hull — 4 min

- Sketch a wide sealed hull.
- Mark one likely weak point.

4 Step 4: Shape the foil — 8 min

- Fold the edges above the water line.
- Press the bottom flat.
- Use no extra foil.

5 Step 5: Float and load — 8 min

- Place the hull on the water.
- Add one cargo piece at a time.
- Record the last safe total.

6 Step 6: Reshape the hull — 5 min

- Change one hull feature.
- Repeat the load test.

7 Step 7: Dry the station — 2 min

- Remove every cargo piece.
- Dry the table.

8 Step 8: Exit check — 2 min

- Write the better cargo total.

What to turn in

Your best cargo count and how hull shape helped.

No computer? Another way to do it

Dry version: fold hulls and predict while the teacher floats one.

Finished early? Next challenge

Cargo Deck: Create a removable paper loading guide that helps another team balance cargo evenly.

Stay safe

Keep water inside tubs, wipe spills immediately, and keep electronics out of the wet zone.

Grade targets

Grade 5: Count cargo with tally marks.
Grade 6: Compare two hull shapes and explain stability.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.